

AI-Powered UTI Antibiotic System

ANALYSIS REPORT • ACADEMIC RESEARCH DOCUMENT

Patient Demographics

Age / Gender	34.0 / Female
Department	General Medicine
Diagnosis	Urethritis
UTI Classification	Recurrent UTI
Sample Type	Urine

Clinical Presentation

Chief Complaints: Burning urination;Urgency

Comorbidities: None

Surgical History: None

Social History: Non smoker

Laboratory Results

Test Name	Value	Unit	Normal Range
Cbp Lymphocytes	38.0	%	20 - 40
Wbc	9000.0	cells/ μ L	4000 - 11000
Polymorphs	64.0	%	40 - 75
Crp	9.0	mg/L	< 10
Rft Serum Creatinine	0.7	mg/dL	0.6 - 1.3
Serum Uric Acid	4.1	mg/dL	3.5 - 7.2
Blood Urea	25.0	mg/dL	7 - 20
Cue Pus Cells	12.0	/HPF	0 - 5
Epithelial Cells	3.0	/HPF	0 - 5
Proteins	Trace		Negative
Rbc	1.0	/HPF	0 - 3

AI Pathogen Prediction

Predicted Organism	Streptococcus agalactiae
Bacteria Type	Gram-positive coccus (Group B Streptococcus; significant in pregnancy)

Antibiotic Susceptibility Profile

Predicted Sensitive	Daptomycin
Predicted Resistant	Azithromycin

Recommended Therapy

Daptomycin

Dosage: 6 mg/kg IV once daily (adjusted to 5 mg/kg if CrCl 30-50 mL/min; max 10 mg/kg).

Precautions: Monitor serum creatinine and creatinine clearance; adjust dose in renal impairment. Check CPK twice weekly to screen for myopathy, especially with concomitant statins or other myotoxic drugs. Avoid use in patients with known hypersensitivity to lipopeptide antibiotics.

Rationale: Daptomycin is active against Streptococcus agalactiae and is listed as sensitive in the prediction. It provides reliable bactericidal activity for Group B streptococci, including strains resistant to macrolides such as azithromycin. The drug is administered intravenously and is appropriate for serious lower urinary tract infections caused by this organism.

Antibiotic Drug History

Clinical Summary

Clinical Summary

The patient, a 34-year-old female with recurrent urethritis, is infected with **Streptococcus agalactiae (Group B Streptococcus)**. The organism shows **macrolide resistance** (azithromycin) and is **susceptible to daptomycin**.

Recommended Therapy

- **Daptomycin 6 mg/kg IV once daily** (adjust to 5 mg/kg if CrCl 30-50 mL/min; max 10 mg/kg).
- *Rationale:* Daptomycin is bactericidal against GBS, including macrolide-resistant strains, and is the only antibiotic in the prediction set that remains effective. Intravenous administration is appropriate for a serious lower urinary-tract infection.

Key Precautions

1. **Renal monitoring** - track serum creatinine and CrCl; dose adjust in renal impairment.
2. **Myopathy surveillance** - check CPK twice weekly, especially if the patient is on statins or other myotoxic agents.
3. **Allergy check** - avoid in patients with known hypersensitivity to lipopeptide antibiotics.

This regimen addresses the resistant profile, offers reliable coverage for GBS urethritis, and includes essential safety monitoring.